

STUDENT ATTENDANCE MANAGEMENT SYSTEM

AI23431- WEBTECHNOLOGY AND MOBILE

APPLICATION

MINI-PROJECT REPORT

SUBMITTED BY

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in partial fulfillment for the award of the degree of

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in

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE

AND DATA SCIENCE

RAJALAKSHMI ENGINEERING COLLEGE

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CHENNAI-602 105

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RAJALAKSHMI ENGINEERING COLLEGE

(an Autonomous Institution Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

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MUKESH . D(241801173)

DEPARTMENT VISION

- To promote highly Ethical and Innovative Computer Professionals through excellence in teaching, training and research.

DEPARTMENT MISSION

- To produce globally competent professionals who are motivated to learn emerging technologies and demonstrate innovation in solving real-world problems.
- To promote research activities among students and faculty members that contribute meaningfully to society.
- To instill strong moral and ethical values in professional practice.

PROGRAMME EDUCATIONAL OBJECTIVES(PEO'S)

PEO 1: To equip students with essential background in computer science, basic electronics and applied mathematics.

PEO 2: To prepare students with fundamental knowledge in programming languages, and tools and enable them to develop applications.

PEO 3: To encourage the research abilities and innovative project development in the field of AI, ML, DL, networking, security, web development, Data Science and also emerging technologies for the cause of social benefit.

PEO 4: To develop professionally ethical individuals enhanced with analytical skills, communication skills and organizing ability to meet industry.

PROGRAMME OUTCOMES (POs)

PO 1: Engineering knowledge: Apply the knowledge of Mathematics, Science, Engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO 2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO 3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO 4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO 5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO 6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO 9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO 11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO 12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

A graduate of the Artificial Intelligence and DATA SCIENCE Program will demonstrate

PSO 1: Foundation Skills: Ability to understand, analyze and develop computer programs in the areas related to algorithms, system software, web design, AI, machine learning, deep learning, data science, and networking

PSO 2: Problem-Solving Skills: Ability to apply mathematical methodologies to solve computational task, model real world problem using appropriate AI and ML algorithms. To understand the standard practices and strategies in project development, using open-ended programming environments to deliver a quality product.

PSO 3: Successful Progression: Ability to apply knowledge in various domains to identify research gaps and to provide solution to new ideas, inculcate passion towards higher studies, creating innovative career paths to be an entrepreneur and evolve as an ethically social responsible AI and DS professional.

COURSE OBJECTIVE

- To provide foundational knowledge and practical skills in creating and structuring web pages using HTML, enabling students to build accessible and well-organized websites.
- To understand and practice Embedded Dynamic Scripting on Client-side Internet Programming.
- To implement Server Side Scripting.
- To facilitate students to understand android Application Design
- To help students to gain a basic understanding of Android application development

COURSE OUTCOME

On completion of course students will be able to,

- CO 2: Upon completing the course, students will be able to design and develop basic web pages with proper structure and semantic elements, ensuring accessibility and functionality.
- CO 3: Design and implement dynamic web page with validation using javascript objects and by applying different event handling mechanism.
- CO 4: Design and implement simple webpage to learn JSP and Servlet.
- CO 5: Design and implement simple Application Design.
- CO 6: Identify various concepts of mobile programming that make it unique from programming for other

CO-PO-PSO Mapping

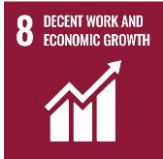
CO	PO 1	P O 2	P O 3	P O 4	P O 5	P O 6	P O 7	P O 8	P O 9	P O 10	P O 11	P O 12	PS O 1	PS O 2	PS O 3
CO 1	3	3	3	3	3	2	2	2	3	2	3	3	3	3	3
CO 2	3	3	3	3	3	2	-	-	2	2	2	3	3	2	2
CO 3	3	3	3	2	3	1	1	2	3	3	3	3	3	3	3
CO 4	3	3	3	3	3	2	1	2	3	2	2	3	3	3	3
CO 5	1	1	1	1	1	-	-	-	3	3	3	3	1	-	2

Note: Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3:

Substantial (High) No correlation: “-”

SUSTAINABLE DEVELOPMENT GOALS

SDG	GOAL	JUSTIFICATION
 SDG-4	Quality Education	Education drives progress on other SDGs, breaks poverty cycles, and promotes peace and inequality reduction.
 SDG-8	Decent Work and Economic Growth	Necessary for social stability, shared prosperity, and combating poverty.
 SDG-9	Industry, Innovation, and Infrastructure	Infrastructure is vital for economic growth and stability. It acts as a poverty reduction tool through job creation and provides technological solutions to environmental challenges.
 SDG-17	Partnerships for the Goals	Given the interconnected nature of the SDGs, success requires collaborative action, global resource sharing, and bridging the immense financing gap faced by developing nations.

ABSTRACT

The Online Restaurant Table Booking System is a web-based application developed to simplify and automate the process of restaurant table reservations. The system provides customers with an easy and convenient platform to book tables online without visiting the restaurant physically or making phone calls. It helps restaurants manage reservations efficiently while improving customer satisfaction and reducing manual workload.

The application allows users to register and log in securely, browse restaurant details, view available tables and seating capacity, select preferred booking dates and time slots, and confirm reservations digitally. The system displays table availability in real time using an interactive seat-selection interface similar to theatre seat booking systems. Customers can choose the number of seats required and reserve suitable tables based on availability.

The system also includes an OTP-based booking verification mechanism to enhance security and avoid fake reservations. After confirming the booking, users receive a One-Time Password (OTP) through email for verification. Once verified, the booking is successfully confirmed, and a confirmation email containing reservation details is sent to the customer automatically.

An admin dashboard is provided for restaurant administrators to manage bookings, monitor table occupancy, update slot availability, and handle customer reservations efficiently. The system reduces paperwork, minimizes booking conflicts, and improves operational efficiency for restaurant management.

The frontend of the application is developed using React.js and Tailwind CSS to provide a modern and responsive user interface. The backend is implemented using Node.js and Express.js, while MongoDB is used for database management. Email notifications and OTP verification are handled using Nodemailer integration.

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- **LIST OF ABBREVIATIONS**

- STPS – Smart Travel Planning
System DB – Database
- SQL – Structured Query Language
- API – Application Programming
Interface UI – User Interface
- HTML – HyperText Markup
Language JS – JavaScript
- HTTP – HyperText Transfer
Protocol SMTP – Simple Mail
Transfer Protocol CRUD – Create,
Read, Update, Delete

CHAPTER 1

INTRODUCTION

1.1 General

In today's fast-growing digital world, online services have become an important part of everyday life. People prefer online platforms because they provide convenience, speed, accuracy, and easy accessibility. Restaurants and food businesses are also adopting digital technologies to improve customer experience and simplify reservation management. Traditional restaurant reservation methods such as phone calls and walk-in bookings often create confusion, long waiting times, and poor table management. To overcome these issues, an Online Restaurant Table Booking System is developed.

The Online Restaurant Table Booking System is a web-based application designed to automate restaurant table reservation processes. The system enables customers to book restaurant tables online based on available seating capacity, preferred dates, and time slots. It provides an interactive interface similar to theatre seat booking systems where users can visually select available tables and reserve them instantly.

The application allows users to register, log in securely, view restaurant details, browse menus, check table availability, and confirm bookings digitally. The system also includes OTP-based verification for secure reservation confirmation. After successful booking, customers receive confirmation emails containing reservation details such as booking ID, table number, booking date, and timing.

1.2 Need for Online Restaurant Table Booking

The need for the Online Restaurant Table Booking System arises because traditional restaurant reservation methods are not efficient enough to handle modern customer expectations. Customers require fast, secure, and convenient booking facilities that can be accessed digitally.

The major needs for the system are:

1. Online Reservation Facility

Customers need the ability to reserve restaurant tables online without physically visiting the restaurant.

2. Real-Time Table Availability

The system should display available and reserved tables dynamically to avoid confusion and duplicate bookings.

3. Better Table Management

Restaurants need efficient table allocation systems to optimize seating arrangements and improve service quality.

4. Reduced Waiting Time

Customers can pre-book tables and avoid long waiting queues during busy hours.

5. Secure Booking Verification

OTP-based verification ensures secure and genuine reservations.

6. Automated Booking Confirmation

The system automatically sends confirmation messages and emails to customers after successful reservations.

7. Improved Customer Experience

Modern digital booking systems increase convenience and customer satisfaction.

1.3 Scope of the System

The scope of the Online Restaurant Table Booking System includes the development of a web application for managing restaurant reservations digitally. The system covers functionalities such as customer registration, table booking, seat selection, payment processing, and reservation management.

The project scope includes:

Customer account management

Online table booking

Real-time slot management

Interactive table visualization

OTP verification system

Payment gateway integration

Booking history management

Admin reservation management

The system can be implemented in restaurants, cafes, hotels, and food service businesses to improve reservation management and customer experience.

1.4 Organization of the Report

The project report for the Online Restaurant Table Booking System is organized into seven chapters. Each chapter explains different aspects of the project including system analysis, design, implementation, testing, and future enhancements. The organization of the report is described below:

Chapter 1 – Introduction

This chapter provides an overview of the Online Restaurant Table Booking System. It explains the background of the project, need for the system, existing manual reservation process, drawbacks of the existing system, proposed solution, objectives, scope, advantages, and overall project overview.

Chapter 2 – Literature Survey

This chapter discusses the study and analysis of existing restaurant reservation systems and related online booking platforms. It explains previous research work, technologies used in similar systems, advantages and limitations of existing applications, and the need for improved digital reservation management systems.

Chapter 3 – Problem Formulation and Objectives

This chapter defines the problem statement and explains the issues faced in traditional restaurant reservation systems. It also describes the objectives and goals of the proposed Online Restaurant Table Booking System.

Chapter 4 – System Design

This chapter explains the overall system architecture, database design, data flow, UML diagrams, and system requirements. It also describes the frontend and backend design of the application along with hardware and software requirements.

Chapter 5 – System Implementation

This chapter explains the implementation details of each module in the system. It includes user registration, login, table selection, slot booking, OTP verification, billing, payment processing, admin dashboard, notification system, and database implementation. The technologies and frameworks used for development are also discussed.

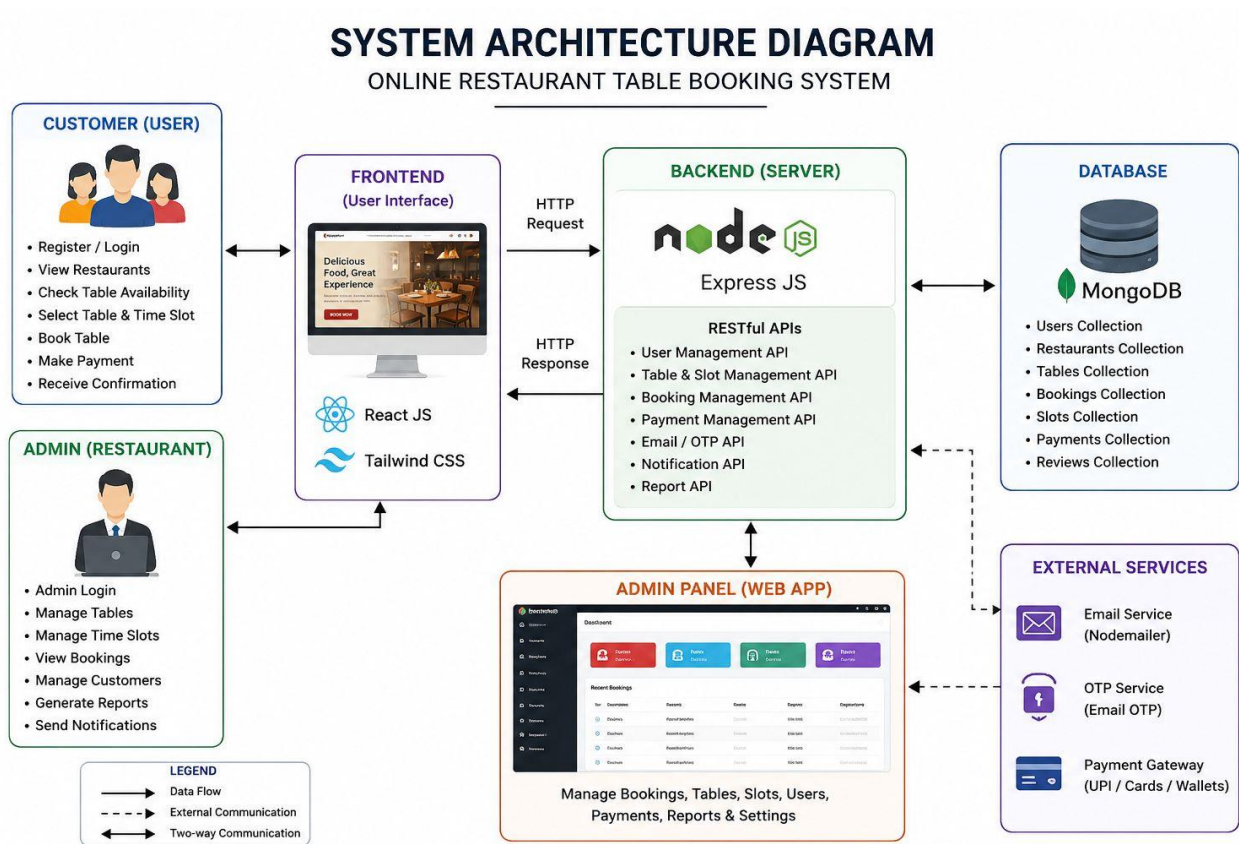
Chapter 6 – Results and Discussion

This chapter presents the outputs and results obtained after implementing the system. It explains system performance, testing methods, screenshots of modules, performance evaluation, and discussion of the overall functionality and efficiency of the application.

Chapter 7 – Conclusion and Future Enhancements

This chapter summarizes the overall project work and concludes the effectiveness of the Online Restaurant Table Booking System. It also discusses future improvements and additional features that can be implemented, such as mobile applications, AI-based table recommendations, real-time analytics, and advanced payment systems.

System architecture diagram:



CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

The literature survey is an important part of software project development because it provides an understanding of existing systems, technologies, methodologies, and research related to the proposed project. It helps identify the strengths and weaknesses of current restaurant reservation systems and provides insights into improving the efficiency and functionality of the proposed Online Restaurant Table Booking System.

With the rapid growth of internet technologies and digital services, many restaurants and food service businesses have started adopting online reservation systems to improve customer convenience and operational management. Traditional reservation methods such as phone calls and walk-in bookings are gradually being replaced by web-based and mobile-based reservation platforms. These modern systems help restaurants manage customer bookings efficiently, reduce waiting times, and optimize table utilization.

Several online restaurant booking platforms such as OpenTable, Zomato Dining, EazyDiner, and TableAgent provide digital reservation services. These systems allow customers to search restaurants, check table availability, reserve tables, and receive booking confirmations online. Some systems also provide additional features such as online payments, digital menus, customer reviews, and loyalty programs.

2.2 Existing Online Restaurant Table Booking

The existing restaurant reservation system mainly depends on traditional manual booking methods such as phone calls, direct walk-in reservations, and handwritten booking registers. In many restaurants, customers contact the restaurant staff through telephone calls to reserve tables. The staff manually checks table availability and records reservation details in notebooks, spreadsheets, or simple management systems.

In some restaurants, reservation details are maintained using basic software applications without real-time synchronization and automation features. These systems often lack proper table visualization, dynamic slot management, automated confirmations, and secure booking verification. Customers may face difficulties in checking table availability instantly, especially during peak hours, weekends, and special occasions.

Modern online reservation platforms such as OpenTable, Resy, and EazyDiner provide digital booking facilities for restaurants. These platforms allow customers to

reserve tables online and receive booking confirmations. However, many small and medium-sized restaurants still do not use advanced reservation systems due to high operational costs and complexity.

The existing systems have several limitations:

1. Manual Reservation Process

Most restaurants still depend on manual reservation methods, which consume time and increase workload for staff members. Manual systems are more prone to human errors such as duplicate bookings and incorrect reservation entries.

2. Lack of Real-Time Table Availability

Traditional reservation systems do not provide real-time table status updates. Customers cannot instantly know whether tables are available or already reserved. This may result in confusion and booking conflicts.

3. Poor Customer Experience

Customers often experience long waiting times and difficulty contacting restaurants during busy hours. Lack of instant confirmation and delayed communication reduce customer satisfaction.

4. No Interactive Table Selection

Most existing systems do not provide visual table selection interfaces. Customers cannot choose tables based on seating arrangements or preferences.

5. Increased Human Errors

Manual reservation handling may lead to incorrect bookings, missing entries, or overlapping reservations. Such issues create inconvenience for both customers and restaurant staff.

2.3 Related Work

Several researchers and software developers have worked on online restaurant reservation systems and digital table management applications to improve restaurant operations and customer convenience. Existing research and commercial applications mainly focus on reservation automation, online booking facilities, customer management, and digital payment integration.

Many modern restaurant management systems use web technologies and database systems to simplify table booking and reservation handling. These systems help reduce manual work, improve booking accuracy, and optimize restaurant seating arrangements.

1. Online Restaurant Reservation Systems

Existing online reservation platforms such as OpenTable, Resy and EazyDiner provide customers with online table reservation facilities. These platforms allow users to search restaurants, check availability, reserve tables, and receive booking confirmations digitally.

Research studies show that digital reservation systems improve customer satisfaction and reduce waiting times significantly. Restaurants using online reservation platforms can manage customer flow more efficiently and increase operational productivity.

2. Web-Based Restaurant Management Applications

Several web-based restaurant management systems have been developed using technologies such as PHP, JavaScript, React.js, Node.js, and MySQL/MongoDB databases. These applications mainly focus on:

- Table reservation management
- Customer information storage
- Food ordering systems
- Billing and payment processing
- Reservation tracking

Most systems use client-server architecture where frontend applications communicate with backend servers and databases to process booking requests. Modern systems also include responsive user interfaces and mobile compatibility.

3. Digital Booking and Scheduling Systems

Research related to online appointment scheduling systems in healthcare, hotels, and restaurants shows that digital booking platforms improve resource utilization and reduce scheduling conflicts. Real-time slot management and automated notifications are considered important features for improving booking efficiency.

Many scheduling systems use:

- Real-time database updates
- Automated notifications
- Email confirmations
- OTP verification systems
- Interactive booking interfaces

These features help prevent duplicate reservations and improve user trust.

2.4 Summary

The literature survey provides a detailed study of existing restaurant reservation systems, online booking platforms, and related web-based management applications. The survey explains how digital reservation systems have improved restaurant operations by automating booking processes, reducing waiting times, and enhancing customer convenience.

Existing platforms such as OpenTable, Resy and EazyDiner provide online reservation and table management facilities. Research studies and existing applications demonstrate that online booking systems improve operational efficiency, optimize table utilization, and provide better customer experiences compared to traditional manual reservation methods.

CHAPTER 3

PROBLEM FORMULATION AND OBJECTIVES

3.1 Problem Statement

In today's fast-paced digital environment, customers expect quick, convenient, and reliable online services in every sector, including restaurant reservation systems. However, many restaurants still depend on traditional manual reservation methods such as phone calls, walk-in bookings, and handwritten reservation records. These methods are inefficient, time-consuming, and difficult to manage, especially during peak hours, weekends, and special occasions.

In the existing manual reservation system, customers often face difficulties in checking table availability and confirming bookings. Restaurants may experience reservation conflicts, duplicate bookings, poor table management, and communication delays. Customers are also required to wait for long periods due to inefficient reservation handling and lack of proper scheduling systems.

Many restaurants do not provide real-time table availability information, making it difficult for customers to know whether seats are available before visiting the restaurant. In addition, the absence of interactive table selection systems prevents customers from choosing tables according to their seating preferences. Traditional systems also lack automated booking confirmation and secure verification mechanisms, increasing the possibility of fake or incorrect reservations.

Another major issue in existing systems is the lack of centralized reservation management. Restaurant staff must manually manage customer details, booking records, table allocation, and reservation schedules, which increases workload and chances of human error. Maintaining paper-based records is also inefficient and difficult to manage for large numbers of customers.

Existing restaurant booking systems may also have limitations such as:

Lack of real-time reservation updates

No visual table selection interface

Poor customer communication

Increased manual workload

Limited online payment facilities

No automated email notifications

Poor reservation tracking and management

3.2 Objectives of the System

The main objective of the Online Restaurant Table Booking System is to develop a modern web-based application that automates restaurant reservation management and provides customers with an easy, secure, and convenient online table booking experience. The system aims to improve restaurant operations, reduce manual work, and enhance customer satisfaction through digital reservation services.

The specific objectives of the system are as follows:

1. To Provide Online Table Booking Facility

The system allows customers to reserve restaurant tables online without visiting the restaurant physically or making phone calls. Customers can access the booking system anytime and from anywhere using internet-enabled devices.

CHAPTER 4

SYSTEM DESIGN

4.1 Introduction

System design is an important phase in software development that defines the overall architecture, components, modules, interfaces, and data flow of a system. It acts as a bridge between the problem analysis and the actual implementation of the project. The main purpose of system design is to convert user requirements into a structured and organized solution that can be efficiently developed and maintained.

In the Online Restaurant Table Booking System, the system design phase focuses on creating an easy-to-use platform that allows customers to search restaurants, check table availability, and reserve tables online. It also helps restaurant administrators manage bookings, customer details, seating arrangements, and reservation schedules effectively. The system design includes several important modules such as:

User Registration and Login Module
Restaurant Management Module
Table Booking Module
Payment Module
Notification and Confirmation Module
Admin Management Module

4.2 System Architecture

The system architecture of the Online Restaurant Table Booking System represents the overall structure of the application and shows how different components interact with each other. The architecture is designed to provide efficient communication between users, administrators, the application server, and the database system.

- **Presentation Layer:**

This layer represents the user interface where users and administrators interact with the system. It includes web pages such as login pages, dashboards, travel planning pages, and itinerary views developed using HTML, CSS, and JavaScript.

- **Application Layer:**

This layer handles the business logic of the system. It processes user requests, verifies login credentials, manages travel plans, generates itineraries, and communicates with the

database. The backend of the system is implemented using Node.js.

- **Database Layer:**

This layer stores all the system data including user details, travel destinations, itineraries, bookings, and travel history. The database ensures secure storage and easy retrieval of information when required.

This architecture ensures smooth communication between users and the database while maintaining system efficiency and reliability.

4.3 System Requirements

System requirements define the hardware and software resources needed to run the application successfully.

4.3.1 Hardware Requirements

The Smart Travel Planning System does not require high-end hardware and can run on standard computers.

- Processor: Intel Core i3 or higher
- RAM: Minimum 4 GB
- Storage: 500 GB Hard Disk or SSD
- Input Devices: Keyboard and Mouse
- Internet Connectio

4.3.2 Software Requirements

The system is developed using commonly available software tools and technologies

- Operating System: Windows 10 or later
- Frontend Technologies: HTML, CSS, JavaScript
- Backend Technology: Node.js
- Database: MySQL
- Development Environment: Visual Studio Code
- WebBrowser: Google chrome

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 System Methodology

The system methodology describes the approach and techniques used for the development and implementation of the Online Restaurant Table Booking System. It explains how the system is planned, designed, developed, tested, and deployed to ensure efficient performance and user satisfaction.

The Online Restaurant Table Booking System follows a structured software development methodology to create a reliable and user-friendly application. The methodology ensures proper communication between different modules and guarantees that the system meets the requirements of customers and restaurant administrators.

Development Methodology

The project follows the Waterfall Model methodology because it provides a systematic and step-by-step approach for software development. In this model, each phase is completed before moving to the next phase.

The major phases involved are:

Requirement Analysis

System Design

System Development

Testing

Deployment

Maintenance

5.2 Module Descriptions

The Online Restaurant Table Booking System is divided into several modules to simplify the development process and improve system efficiency. Each module performs a specific function and works together to provide a complete restaurant reservation solution.

1. User Module

The User Module manages all customer-related activities in the system. It allows users to create accounts, log in, search restaurants, and book tables online.

Functions:

User registration

User login and authentication

Profile management

Search restaurants

View table availability

Book and cancel reservations

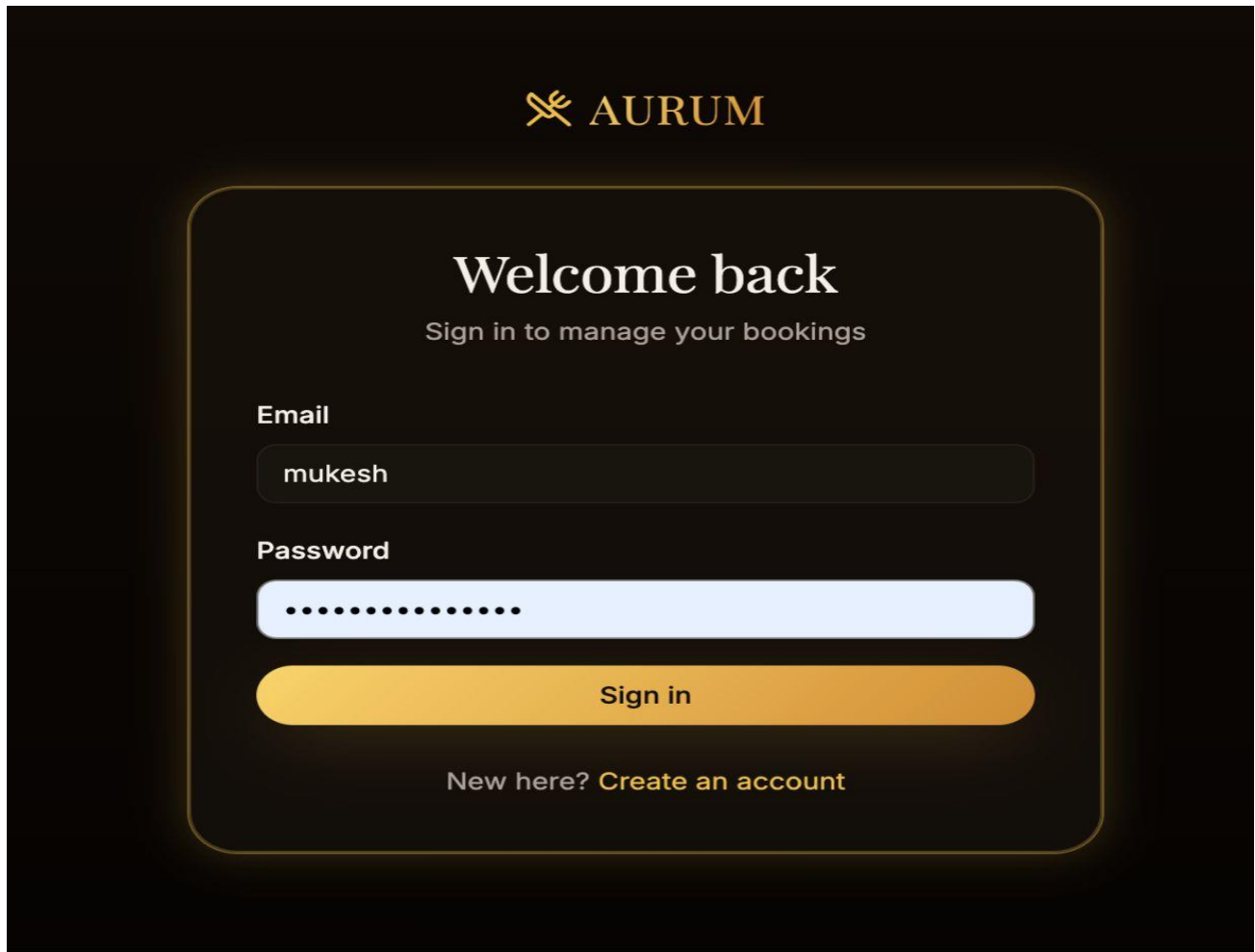
View booking history

Benefits:

Provides a personalized booking experience

Saves customer booking details

Enables easy reservation management



5.3 Landing Page / Gateway Module

The Landing or Gateway Module is the entry point of the Online Restaurant Table Booking System. It provides the first interaction between the user and the application. This module is designed to offer a simple, attractive, and user-friendly interface that helps users easily access different features of the system.

The main objective of the Landing/Gateway Module is to guide users to the appropriate services such as login, registration, restaurant browsing, and booking functionalities.

Features of the Landing / Gateway Module

1. Home Page Display

The module displays the homepage of the application with important information such as:

Restaurant highlights

Featured restaurants

Special offers and discounts

Popular food categories

Customer reviews and ratings

2. User Navigation

It provides navigation menus that allow users to move easily between different sections of the website.

Navigation Options:

Home

Restaurants

About Us

Contact

Login

Register

Booking Page

3. User Authentication Access

The gateway module allows users and administrators to access the system securely through:

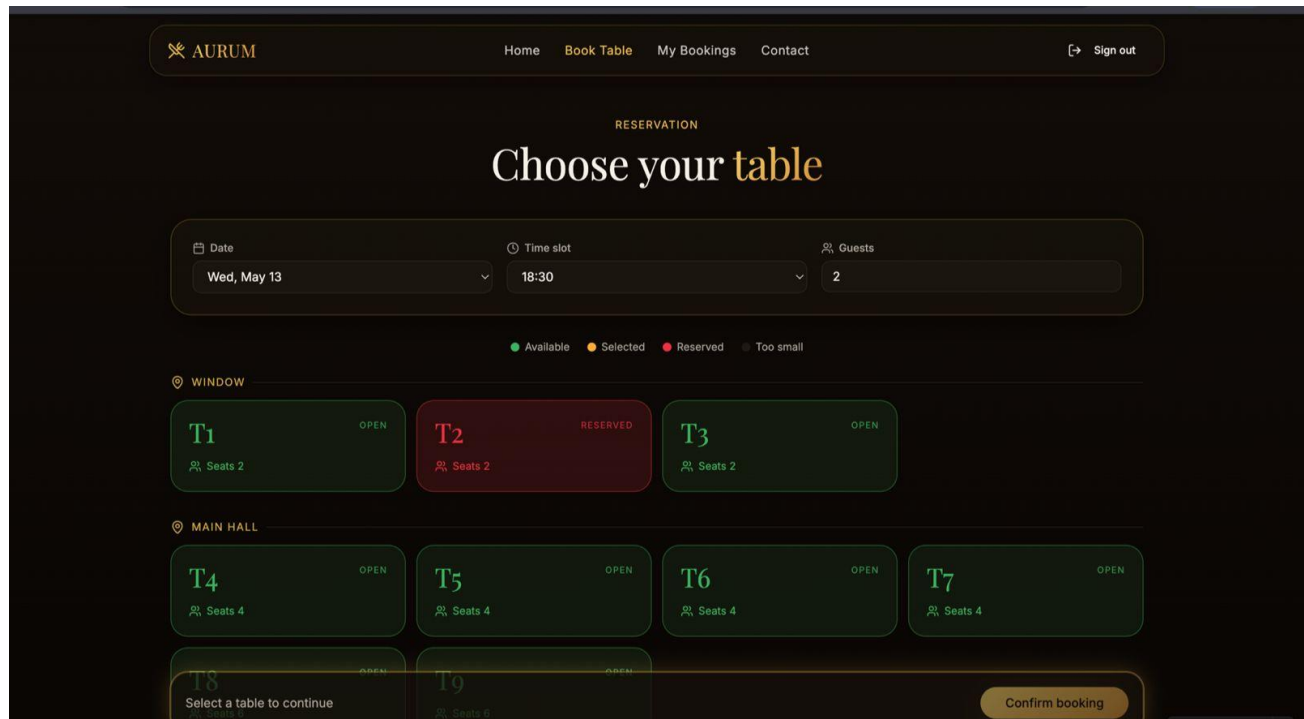
User Login

User Registration

Admin Login

Password Recovery

This ensures that only authorized users can access specific functionalities.



5.2.1 User Registration Module

The User Registration Module is an important component of the Online Restaurant Table Booking System. This module allows new users to create an account in the system by providing their personal details. Registration helps the system identify users and provide personalized booking services.

The main objective of this module is to collect and store user information securely so that customers can access booking features, manage reservations, and receive notifications.

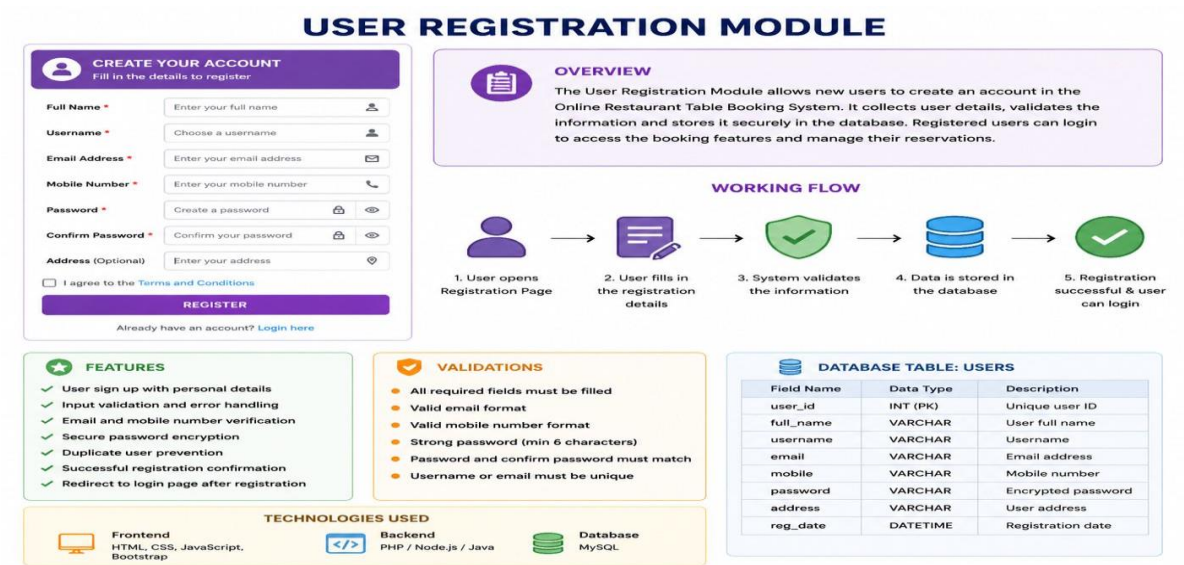
Features of the User Registration Module

1. New User Registration

New users can create an account by entering required details such as:

Full Name
Email Address
Mobile Number
Username
Password
Address (optional)

The system validates the entered details before account creation.



5.2.2 Customer Login Module

The Customer Login Module is an essential part of the Online Restaurant Table Booking System. This module allows registered users to securely access their accounts using valid login credentials such as username/email and password. It ensures authorized access to booking services and protects user information from unauthorized users.

The main objective of the Customer Login Module is to authenticate users and provide secure access to restaurant reservation functionalities.

Features of the Customer Login Module

1. Secure User Authentication

The module verifies user credentials before granting access to the system.

Authentication Details:

Username or Email ID

Password

The system checks whether the entered credentials match the records stored in the database.

2. Password Security

To ensure user account protection, the system uses:

Encrypted passwords

Secure login validation

Session management

Unauthorized access prevention

This improves the overall security of the application.

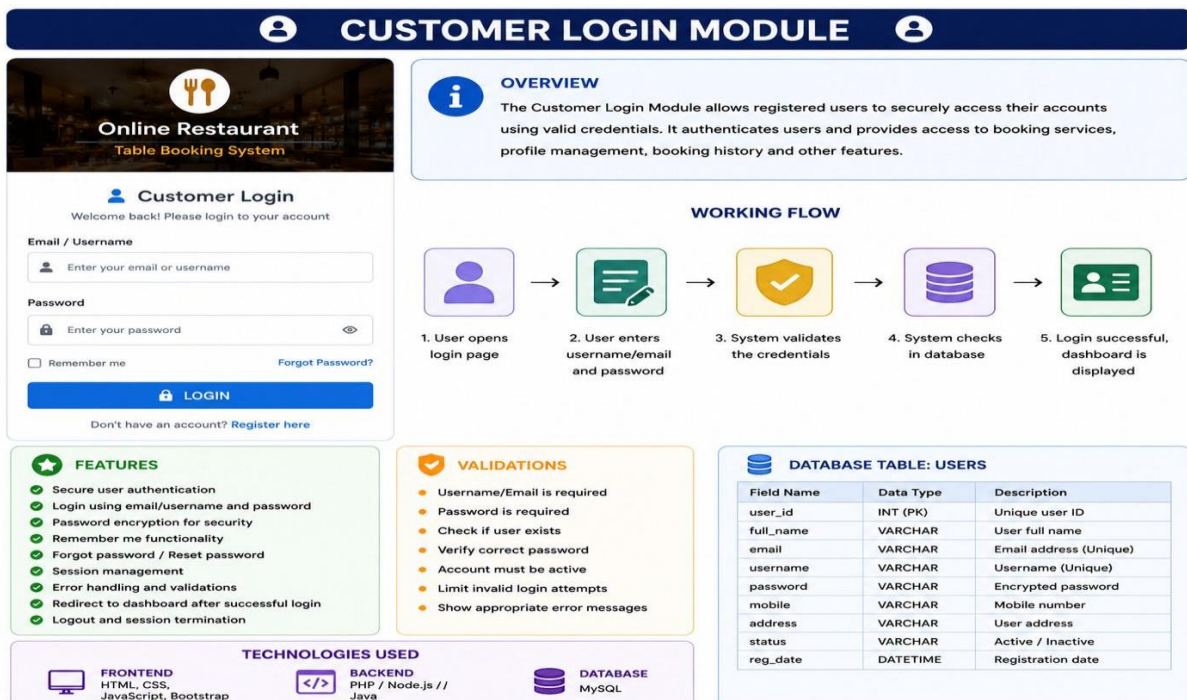
3. Forgot Password Option

Users who forget their passwords can recover or reset them using:

Registered email address

OTP verification

Password reset link



5.2.3 Restaurant Home Page Module

The Restaurant Homepage Module is one of the most important components of the Online Restaurant Table Booking System. It acts as the main interface where customers can explore restaurant details, view available services, and access booking functionalities. The homepage is designed to provide an attractive, user-friendly, and informative experience for customers.

The primary objective of this module is to display restaurant information clearly and help users quickly navigate to table booking and other services.

Features of the Restaurant Homepage Module

1. Restaurant Information Display

The homepage provides complete restaurant details such as:

Restaurant name

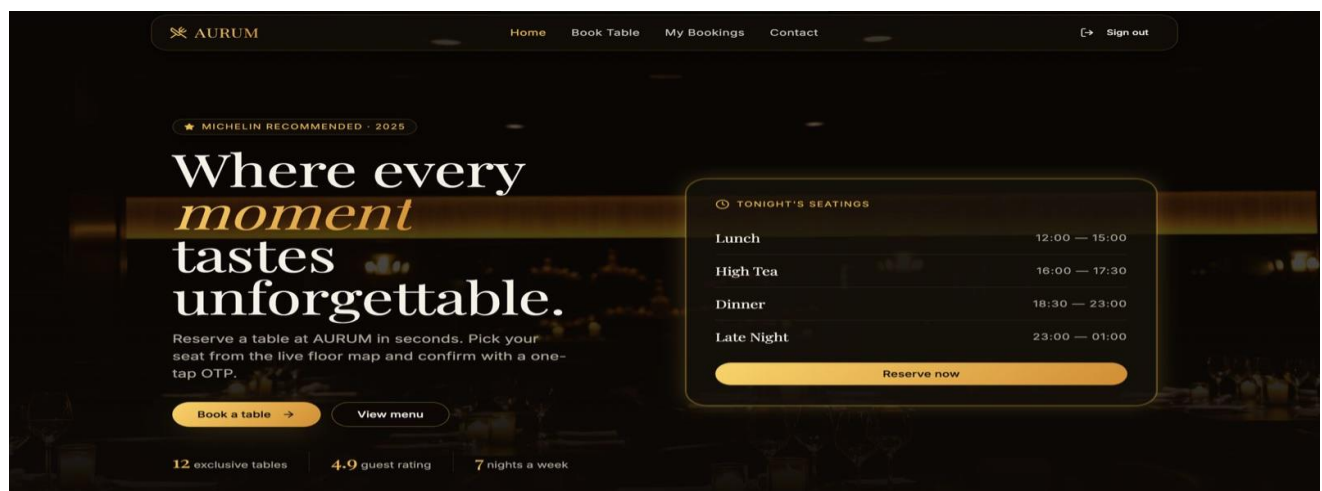
Location and address

Contact information

Operating hours

Restaurant description

.



5.2.3 Table Availability Display Module

The Table Availability Display Module is a vital component of the Online Restaurant Table Booking System. This module allows customers to view the availability of tables in real time before making reservations. It helps users identify vacant tables based on selected date, time, and seating capacity.

The primary objective of this module is to provide accurate table availability information and simplify the reservation process for customers.

Features of the Table Availability Display Module

1. Real-Time Table Availability

The module displays currently available tables based on:

Booking date

Booking time

Number of guests

Seating type

This helps customers quickly identify suitable tables.

2. Table Information Display

Users can view details such as:

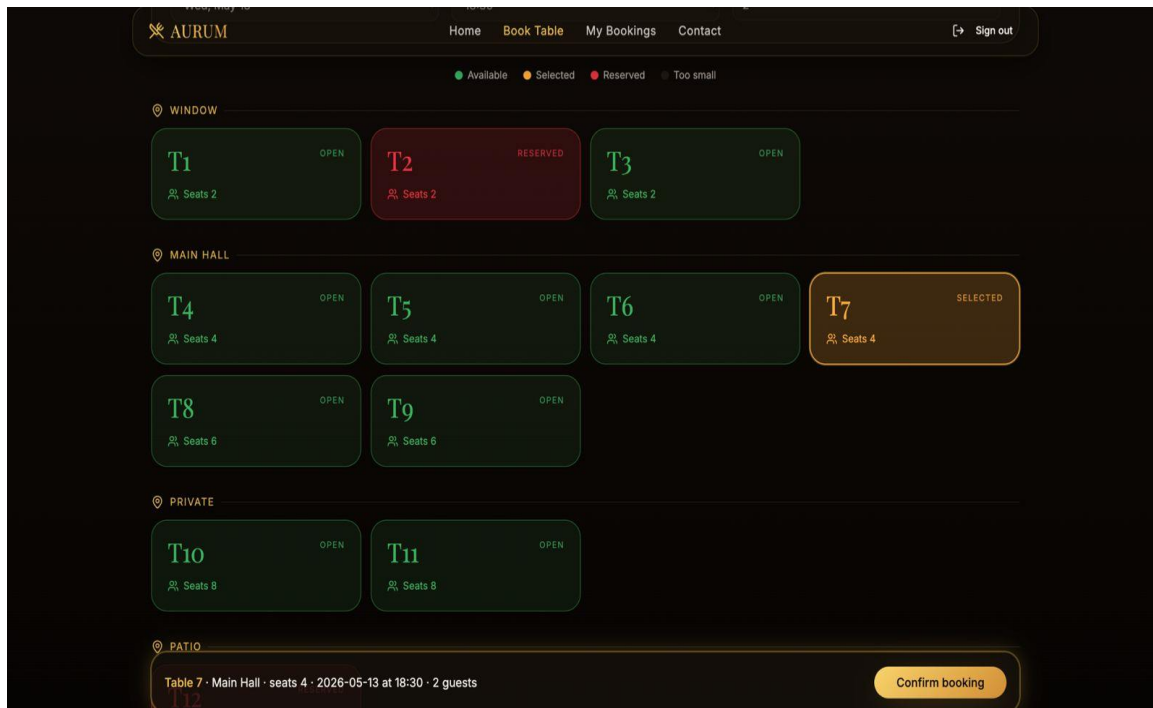
Table number

Seating capacity

Table type (VIP, family, regular)

Availability status

Reservation time slot



5.2.5 Theater Style Table Selection Module

1. Visual Seat/Table Layout

The module displays a graphical layout of restaurant seating arrangements, including:

Table positions

Seating rows

VIP sections

Family seating areas

Indoor and outdoor seating

This allows users to choose seats visually instead of manually selecting tables.

2. Real-Time Seat Availability

The system updates seat availability dynamically

Availability Indicators:

Green – Available

Red – Reserved

Yellow – Temporarily blocked

Gray – Unavailable

This helps users quickly identify free seats.

3. Interactive Table Selection

Customers can:

Click on available tables

View seating capacity

Check table details

Select preferred seating location

This improves user interaction and booking flexibility.

CHAPTER 6

RESULTS AND DISCUSSION

6.1 Results

The Online Restaurant Table Booking System was successfully developed and implemented to provide an efficient and user-friendly platform for restaurant table reservations. The system was tested using different booking scenarios to evaluate its performance, accuracy, reliability, and usability.

The results obtained from the implementation show that the system effectively simplifies the restaurant reservation process for both customers and administrators.

6.2 Performance Evaluation

Performance evaluation is an important process used to measure the efficiency, reliability, speed, and accuracy of the Online Restaurant Table Booking System. The system was tested under different conditions to analyze how effectively it performs booking operations, handles user requests, and manages database transactions.

The evaluation confirmed that the system performs efficiently with fast response times, secure processing, and accurate reservation management.

6.3 Resource Utilization

Resource utilization refers to the efficient use of system resources such as processor, memory storage, network, and database during the operation of the Online Restaurant Table Booking System. Proper resource utilization ensures smooth system performance, faster response time,

and reliable user experience.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

The Online Restaurant Table Booking System was successfully designed and developed to provide an efficient, secure, and user-friendly platform for restaurant table reservations. The system simplifies the traditional reservation process by allowing customers to book tables online quickly and conveniently from anywhere.

The project successfully achieved its main objectives, including user registration and authentication, restaurant browsing, real-time table availability checking, theatre-style seat selection, online booking, payment processing, and notification management. The system also provided effective administrative control for managing restaurants, reservations, users, and reports.

The implementation of real-time table availability and interactive seat selection improved booking accuracy and enhanced customer satisfaction. Secure login authentication and payment processing ensured data protection and reliable transaction handling. The modular design and structured architecture improved system maintainability, scalability, and performance.

7.2 Future Work

The Online Restaurant Table Booking System provides an efficient and reliable platform for restaurant reservations. However, the system can be further improved by adding advanced features and modern technologies to enhance functionality, performance, security, and user experience.

Future enhancements can help the system meet growing customer expectations and support large-scale

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APPENDICES

Appendix-1: source code pseudocode

The pseudocode of the Online Restaurant Table Booking System represents the logical flow of the application. It explains the major functionalities and operations performed by the system in a simplified programming format. The pseudocode helps developers understand the working process before actual implementation.

System Initialization Logic

System Initialization Pseudocode

START

INITIALIZE System

LOAD Configuration Files

CONNECT Database

IF Database Connection Successful THEN

 DISPLAY "Database Connected"

ELSE

 DISPLAY "Database Connection Failed"

 TERMINATE System

END IF

LOAD Application Modules

 LOAD User Module

 LOAD Login Module

 LOAD Restaurant Module

 LOAD Booking Module

 LOAD Payment Module

 LOAD Admin Module

 LOAD Notification Module

CHECK Server Status

IF Server Running THEN

 DISPLAY "Server Active"

ELSE

 DISPLAY "Server Error"

 TERMINATE System

END IF

INITIALIZE Session Management

LOAD Homepage

DISPLAY "System Ready"

STOP

User Login Logic

START

DISPLAY Login Page

INPUT Username or Email

INPUT Password

IF Username or Password is Empty THEN

 DISPLAY "Please Enter All Fields"

 STOP

END IF

CONNECT Database

SEARCH User Details using Username or Email

IF User Exists THEN

 VERIFY Password

 IF Password is Correct THEN

 CREATE User Session

 DISPLAY "Login Successful"

 REDIRECT to User Dashboard

ELSE

DISPLAY "Invalid Password"

END IF

ELSE

DISPLAY "User Does Not Exist"

END IF

CLOSE Database Connection

STOP

Duplicate Prevention Logic

START

INPUT Restaurant ID

INPUT Table ID

INPUT Booking Date

INPUT Booking Time

CONNECT Database

SEARCH Booking Records

WHERE

Table ID = Selected Table

AND Booking Date = Selected Date

AND Booking Time = Selected Time

AND Booking Status = "Confirmed"

IF Matching Booking Found THEN

DISPLAY "Table Already Booked"

STOP

ELSE

TEMPORARILY Lock Selected Table

STORE New Booking in Database

DISPLAY "Booking Successful"

END IF

UNLOCK Table after Confirmation

CLOSE Database Connection

STOP

System Features Implemented

- The Online Restaurant Table Booking System was developed with several important features to provide efficient restaurant reservation management and improve customer convenience. The implemented features help users perform booking operations easily while enabling administrators to manage restaurant activities effectively.
- The system integrates user management, booking processing, payment handling, and real-time table availability functionalities into a single platform.

Appendix 2: System Test Verify

Test Case 1: Table Booking Creation

Result: PASS – plan created successfully

Test Case 2: Duplicate Detection

Result: PASS – duplicate entry prevented

Test Case 3: Table Plan Update Result:

PASS – plan updated correctly **Test**

Case 4: OTP Verification

Result: PASS – OTP sent and verified successfully

Test Case 5: Table Booking Viewing

Result: PASS – users can view travel details

Final System Status

- The Online Restaurant Table Booking System was successfully completed and implemented according to the project requirements. All major modules and functionalities were developed, integrated, tested, and verified to ensure smooth and reliable operation of the system.
- The final system provides a secure, efficient, and user-friendly platform for customers to reserve restaurant tables online and for administrators to manage restaurant operations effectively.

CERTIFICATE OF COMPLETION

Mukesh D

has successfully completed the online course:
Java Servlet Basics and JSP 101

This professional has demonstrated initiative and a
commitment to deepening their skills and advancing
their career. Well done!

12th May 2026

Certificate code : 10218723



Krishna Kumar
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